Prabhakar Jaiswal

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Education

Sardar Vallabhbhai National Institute of Technology, Surat, India Bachelor of Technology, Electrical Engineering	*CGPA*: **8.45/10**	***2019 - 2023***
Saraswati Vidya Mandir Munger, Bihar, India Central Board of Secondary Education (CBSE) | XII Standards	*Aggregate*: **79.8%**	***2018***
Saraswati Vidya Mandir Munger, Bihar, India Central Board of Secondary Education (CBSE) | X Standard	*CGPA*: **9.6/10**	***2016***

Work Experience

Assistant Manager - Electrical Maintenance

Vedanta Aluminium Limited — *Jharsuguda, Odisha* (*Sep 2023- Present*)

- Role: Electrical maintenance of the cast house plant, ensuring optimal operation and reliability of equipment.
- Tools & Technologies:
  - *Programming & Control Systems:* RSLogix 5000, Allen Bradley PLC
  - *Hardware:* ABB 4-axis robots, Sensors, VFD Drives (ABB, Siemens, Allen Bradley)
  - *Software:* RobotStudio
- Key Achievement:
  Spearheaded the successful in-house conversion of a 22kg ingot production line to a 10kg ingot line. This involved:
  - Modifying and updating PLC codes and robot programming.
  - Implementing necessary mechanical and software changes to adapt the process.
  - Ensuring seamless transition and minimal downtime, enhancing production efficiency.

Project trainee R&D [link]

TVS Motor Company — *Bangalore, Karnataka* (*May 2022- July 2022*)

- Worked in the R&D department of TVS Motor on a project entitled **“**Retractable Backrest Seat for Two Wheelers.”
- Developed a working prototype of an adjustable backrest with auto-locking, current protection, Limit switch protection and wireless control over Wi-Fi.

Research Internship [link]

SVNIT Surat — *Surat, Gujrat* (*Jan 2022- May 2022*)

- Worked on the project “3D mapping of construction sites with autonomous quadcopter using mission planner and Qground control.”
- Analysed different stereo cameras: ZedFu, Intel Realsense on NVIDIA Jetson Nano computer
- Created a 3D model of an old SVNIT guest house using Photogrammetry.

Projects

SUN: Surveillance using UAV network [link] — *Team Project* (*Nov 2022 – Present*)

- Developed own flight controller on STM32 chip from scratch with features: Auto-level, Altitude hold, GPS hold, Return to home, wireless camera
- Worked on Swarm operation of 5 Drones, mesh communication done using NRFl01 module, Localisation will be done using AruCo markers and a central camera.

OSEM: Optimising and Saving Energy with Management[link] — *Team Project* (*Jan 2023 – May 2023*)

- Home automation with efficient management of energy consumption and also load forecasting with smart control like touch-based control, human presence detection, and IoT control
- Developed a working prototype PCB board, Website, mobile app, and Azure cloud-based VM for running machine learning model

Quadcopter for 3D mapping of construction sites [link] — *Team Project* (*Jan 2022 – May 2022*)

- Built a quadcopter from scratch with Pixhawk 4 FC, T-motor BLDC motors, 10000Mah 4S lipo battery, Calculations done with ecalc.ch

Simulation of Autonomous Car in ROS2 Foxy Gazebo [link] — *Team Project* (*May 2021 – Jul 2021*)

- Applied lane detection using HSV filter OpenCV, A* algorithm, custom obstacle avoidance algorithm using lidar, Yolo v3 traffic sign recognition trained on a custom dataset

Underwater Autonomous Robot [link] — *Team Project* (*Oct 2020 – Mar 2021*)

- Participated in Robofest 2.0 competition organised by GUJCOST.
- Designed an underwater Robot that can be completely submerged and has 3-DoF.
- Multi-terrain land, water and ice surfaces; wireless control; innovative r-hex mechanism.

Relevant Courses---

- Short-term training program on Embedded C for ARM Cortex M4 microcontroller
- Machine learning by Stanford University (Coursera)
- Convolutional neural network by deeplearning.ai (Coursera)

Technical Strengths---

- **Programming Languages** C++, C, Python
- **Software Packages** ROS, Altium, Proteus
- **libraries** OpenCV, Pandas, Matplotlib, NumPy
- **Computer Graphics** Blender, Adobe Premiere Pro
- **Other** Git, Linux

Academic Achievements---

- First runner-up at UDAAN 1.0 Drone Challenge held at the Indian Institute of Technology (IIT) BHU [link]
- 50K INR assistance for the development of PoC under the Swarm drones category in the GUJCOST Robofest 3.0 [link]
- Received funding for two projects, a) SUN and b) OSEM, from Startup cell SVNIT Surat ASHINE. [link]
- Consolation prize of 50k in the GUJCOST Robofest 2.0 stage 2 and 50k assistance prize in stage 1. [link]
- NXP AIM Online design challenge Top 10 finalists among 600+ participants. [link]
- GKDC EV Concept Challenge 2020 Winner. [link]
- Selected for Young Scientist India 2016, organised by Space Kidz India and participated in Chennai finals. [link]
- CBSE National level science fair 2016 New Delhi Participant, State level winner.

Extracurricular Activities---

- Interested in video editing and 3D animations.
- Senior member at Technical Club Drishti at SVNIT, Surat.
- Interested in Badminton.